

Pathways After High School in Northeast Minnesota

Understanding What Keeps Youth in the Region's Workforce

100-word thesis

This report examines the educational and employment pathways of high school graduates in Northeast Minnesota to understand what drives regional workforce retention. Using linked education and employment records over ten years, it finds that students with strong Career and Technical Education (CTE) experiences and regional postsecondary connections are most likely to remain employed in the region. Those with higher academic achievement or degrees earned outside the region are more likely to leave, while economically disadvantaged students face higher risks of unstable employment. The findings highlight actionable strategies to strengthen local career pipelines and retain young talent in rural Minnesota.

48-word thesis

Data analysis found that students with strong Career and Technical Education experiences and regional postsecondary connections tend to remain employed in the region. Those with higher academic achievement and/or degrees earned outside the region tend to leave, while economically disadvantaged students face higher risks of unstable employment.

Introduction

Across rural Minnesota, and especially in the state's northeast region, workforce shortages have become a defining issue. Employers report persistent vacancies, while demographic shifts—aging populations and youth outmigration—limit the pipeline of new workers. In this context, understanding what happens to high school graduates is essential to regional and state economic planning.

This study explores that question in depth: What factors predict whether young people stay, leave, or become disconnected from meaningful employment after high school?

By following over a decade of graduates through education and employment data, we can move beyond anecdotes to see clear patterns—and identify the points where schools, colleges, and communities can intervene.

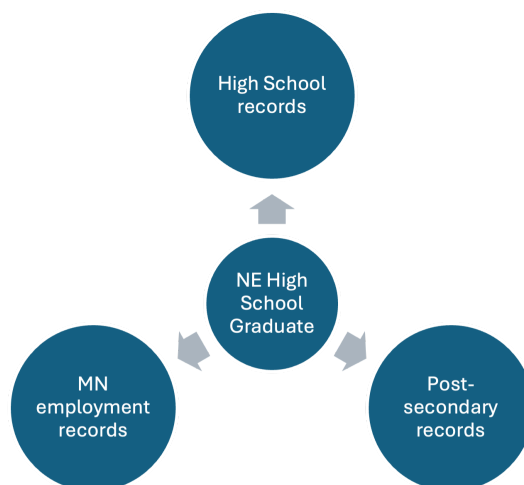
1 - Data Sources and Study Population

This analysis is built on one of the most complete, longitudinal education-to-employment datasets available in Minnesota. It connects individual student records from **high school through college and into the workforce**, providing a rare, data-driven look at what happens after young people graduate.

The study follows every graduate from public high schools in **Economic Development Region 3 (Northeast Minnesota)**—covering communities from the Iron Range to the North Shore—across more than a decade of data. By linking these records, we can see not just *where students go next*, but *how their choices and opportunities unfold over time*.

Three key data systems power this analysis:

- **High School Records:** Include information on each student's course history, participation in **Career and Technical Education (CTE)**, **MCA and ACT scores**, and special populations such as students in special education or qualifying for free/reduced lunch. These data tell us what kind of learning experiences and academic preparation students had before graduation.
- **Postsecondary Records:** Drawn from Minnesota's higher education systems and the **National Student Clearinghouse**, these data capture **college enrollment, persistence, and completion**, both in-state and out-of-state. They reveal which types of colleges students attend—and where those pathways lead.
- **Employment Records:** Provided by the **Minnesota Department of Employment and Economic Development (DEED)**, these files show **where graduates work (in Minnesota), how much they work, and whether their jobs are stable and full-time**. Together, these records allow us to trace employment trends within and beyond the region.



By merging these sources, the study can identify patterns that would otherwise remain invisible—such as the connection between high school CTE participation and long-term local employment, or how college location predicts whether graduates return to the region.

Tracking Outcomes Over Time

Each graduate was followed at three intervals—**1 year, 5 years, and 10 years** after high school—allowing us to observe short-term transitions, mid-career stability, and long-term outcomes.

To make the data comparable, every graduate was assigned to one of five **mutually exclusive categories** at each point in time:

1. **Meaningful Workforce Participation – Northeast (Regional Employment):**
Working 1,000+ hours with one employer within Northeast Minnesota in a calendar year, indicating stable regional employment.
2. **Meaningful Workforce Participation – Minnesota:**
Working 1,000+ hours with one employer elsewhere in the state—typically representing career mobility within Minnesota.
3. **Not Meaningful Workforce Participation:**
Has a MN employment record but does not reach the “meaningful” threshold and is not attending post-secondary. This group typically comprised of individuals working part time.
4. **Attending Postsecondary:**
Enrolled in college but not meeting the threshold for meaningful employment, often reflecting full-time students or those in early transition phases.
5. **No Minnesota Employment Record, Not Attending Postsecondary:**
No employment or enrollment record within Minnesota. This group includes graduates working in other states, in the military or federal service, or temporarily out of the labor force.

A Focus on Persistent Relationships

Rather than capturing a single snapshot, this analysis focuses on **persistent relationships**—factors that remain strong at 1-, 5-, and 10-year intervals. This approach ensures that the patterns described are not temporary or cohort-specific, but **structural and consistent over time**.

In short:

This isn’t a one-year picture—it’s a decade-long look at how education and opportunity interact for thousands of Minnesota graduates.

By following students over time, the study moves beyond surface-level outcomes to uncover the real dynamics of how young people from Northeast Minnesota build—or struggle to build—their working lives after high school.

2 - The Big Picture: Where Do Graduates Go?

Every year, thousands of students walk across a high school graduation stage in Northeast Minnesota—young adults stepping from classrooms into a world of choices: college, work, family, or sometimes uncertainty. Where they go next, and how those choices unfold over time, forms a part of the foundation of the region’s future workforce.

The data show that for many, the first step is **education**. Immediately after graduation, about **half of Northeast Minnesota’s graduates enroll in a postsecondary institution**, whether a local community college, a technical program, or a four-year university. For the rest, high school graduation marks their **entry directly into the workforce**.

But time changes those balances. Each year, college enrollment declines as more graduates transition into work. By around the **seven-year mark**, most have completed their postsecondary journeys and settled into jobs. That’s when we see **meaningful employment peak**—a snapshot of where the region’s young adults ultimately land:

- **30%** are working meaningfully within **Northeast Minnesota**, contributing directly to local industries like healthcare, education, trades, and manufacturing.
- **20%** are employed meaningfully elsewhere in **Minnesota**, often in larger labor markets such as the Twin Cities or Duluth.
- **20%** are working in Minnesota, but are not reaching the “meaningful” workforce participation threshold,
- And a growing share—over **30% by year 10**—have **no Minnesota employment record**, meaning they are likely working out of state, serving in the military, or otherwise outside the reach of Minnesota’s workforce data systems.

Pathways taken by share of graduates X years after graduating high school

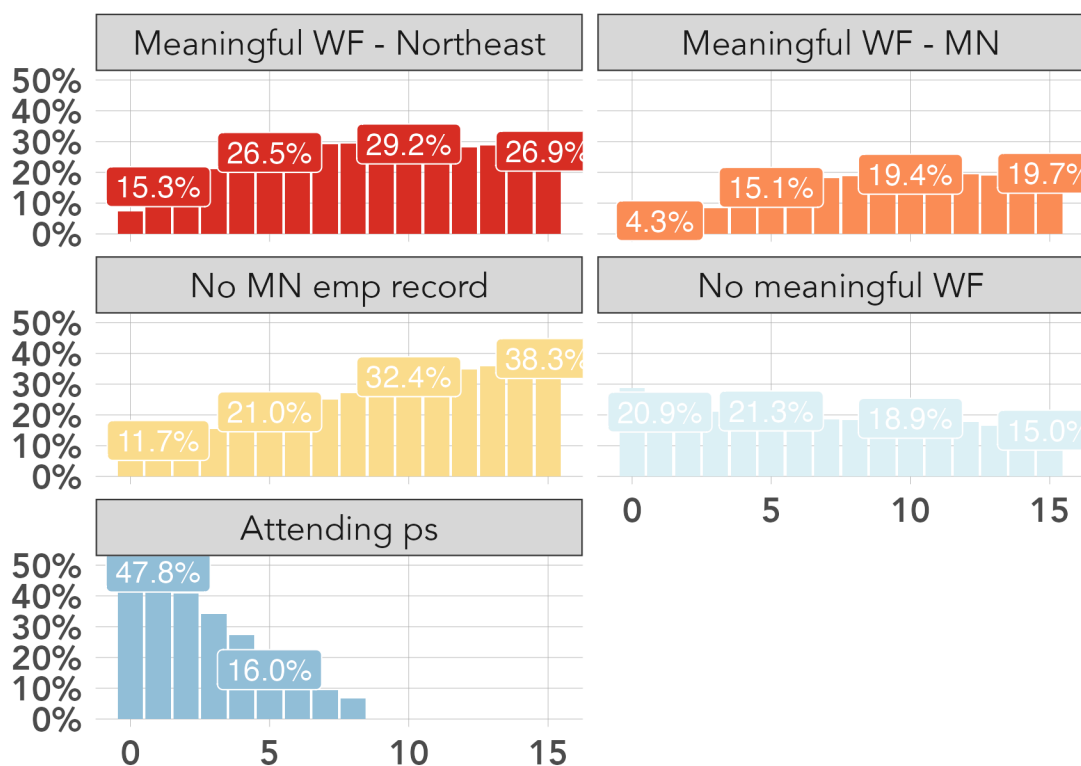


Chart X: As the share of graduates attending post-secondary decreases, the other states of workforce participation increase. The meaningful workforce participation states peak around 7 years after graduating high school while the share of graduates with no MN employment record continues to increase. Data: MN Office of Higher Education - SLEDs

These patterns tell a complex story—one of both **retention and reach**. The region successfully retains a healthy share of its youth, especially through strong local education and training systems. But it also produces graduates whose opportunities pull them outward, toward larger cities and national labor markets. And half of the graduates are a bit disconnected and seem to struggle to find stable footing in work or education.

In other words, Northeast Minnesota's graduates are **not leaving aimlessly—they are following pathways**.

Some are anchored in local opportunities. Others are propelled by academic and career aspirations that take them beyond the region. A few are still searching for connection.

Understanding these pathways—and the factors that predict who stays, who leaves, and who becomes disconnected—is at the heart of our analysis described in the following sections.

3. The Three Pathways Framework: Understanding Life After High School

When we follow Northeast Minnesota’s graduates across a decade of education and employment data, a clear and compelling pattern emerges. Most young people transition into one of **three distinct pathways** after high school — each shaped by opportunity, preparation, and access to support.

These pathways capture the full range of post-high school experiences: those who are anchored in the region to attend local colleges and have meaningful workforce participation in the region, those who leave to pursue higher education and new careers, and those who seem to struggle to find stable footing. Together, they define the story of how Northeast Minnesota’s graduates move from school to work — and where they may fall through the cracks. Obviously an individual can be categorized in any of these throughout the decade of tracking, but the patterns and pathways are quite persistent.

3.1 The Anchoring Pathway: Building Local Lives

This pathway is **rooted in community and practical learning**. Students in this group often participate in **Career and Technical Education (CTE)** courses, apprenticeships, or dual-enrollment programs that expose them to local industries. Many continue their education at **regional community or technical colleges**, earning credentials that directly align with high-demand fields in the region.

Graduates following this route are far more likely to achieve **meaningful employment within Northeast Minnesota**.

Meaningful workforce participation in Northeast

Meaningful workforce participation in the region peaks at nearly 30% of high school graduates

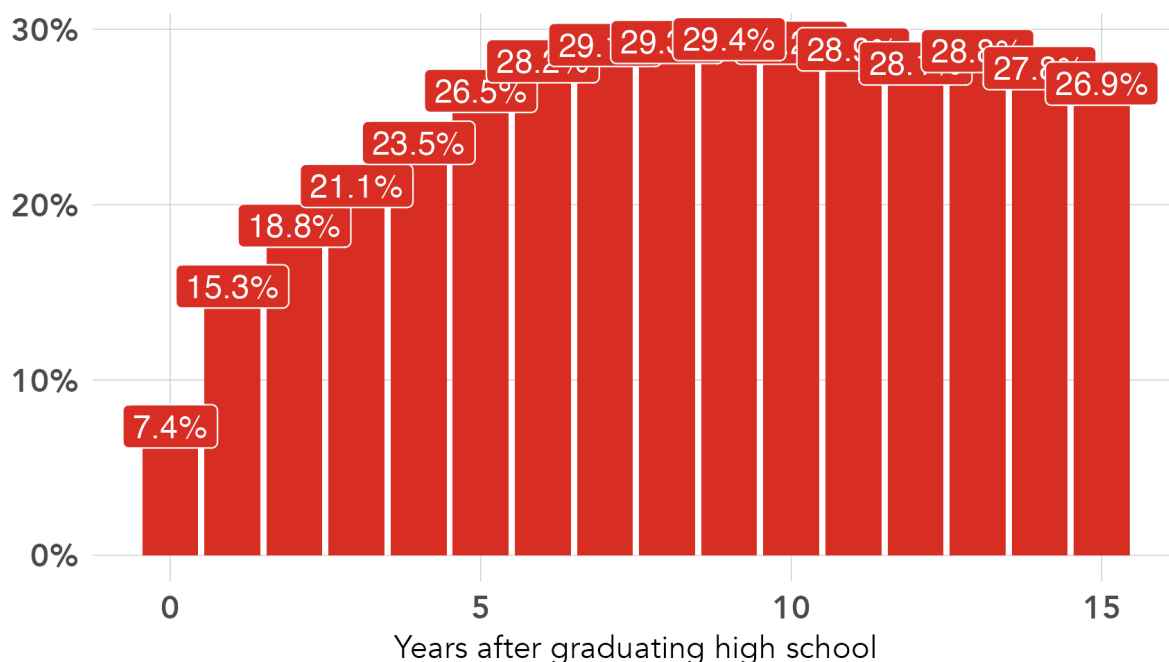


Chart X: The Anchoring Pathway shows how hands-on learning and local postsecondary opportunities keep young people connected to home and invested in their communities.

3.2 The Academic Pathway: Expanding Horizons

This pathway represents **academic readiness and educational mobility**. Students on this route tend to perform well on the **MCA and ACT**, participate in **PSEO or Advanced Placement**, and set their sights on **four-year or out-of-region institutions**.

They pursue advanced degrees, often in fields requiring specialized or professional credentials, and are likely to find **meaningful employment elsewhere in Minnesota or beyond**. They represent the region's contribution to the broader state and national economy — young people whose success stories extend beyond county lines.

Meaningful workforce participation in Minnesota

Meaningful workforce participation in the state peaks at nearly 20% of high school graduates

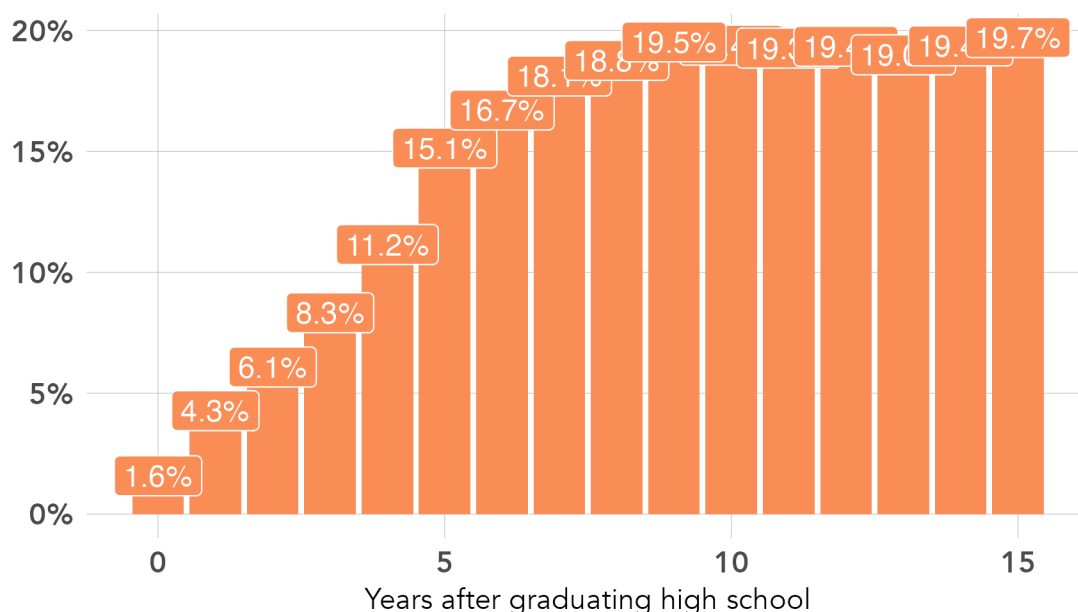


Chart X: The Academic Pathway highlights a positive form of mobility — one driven by preparation and choice — though it can reduce the number of highly educated workers remaining in the region.

3.3 The Unanchored Pathway: Disconnection and Drift

The third pathway represents a **significant share of Northeast Minnesota’s graduates** whose post-high school outcomes are less visible in Minnesota’s data systems. Years after graduation, these individuals often do **not appear in Minnesota’s employment or postsecondary records**, or they are working in jobs below the threshold of “meaningful” employment.

But absence from the data does not necessarily mean absence from the workforce. Many in this group are likely **employed outside Minnesota, serving in the military, or working in federal or self-employed roles** that are not captured by state Unemployment Insurance records. Others, however, are **genuinely disconnected** from both education and stable employment—cycling between temporary jobs, caregiving, or periods out of the labor force.

Workforce participation categories

Graduates that are disconnected from the Minnesota workforce make up nearly half of the high school graduates.

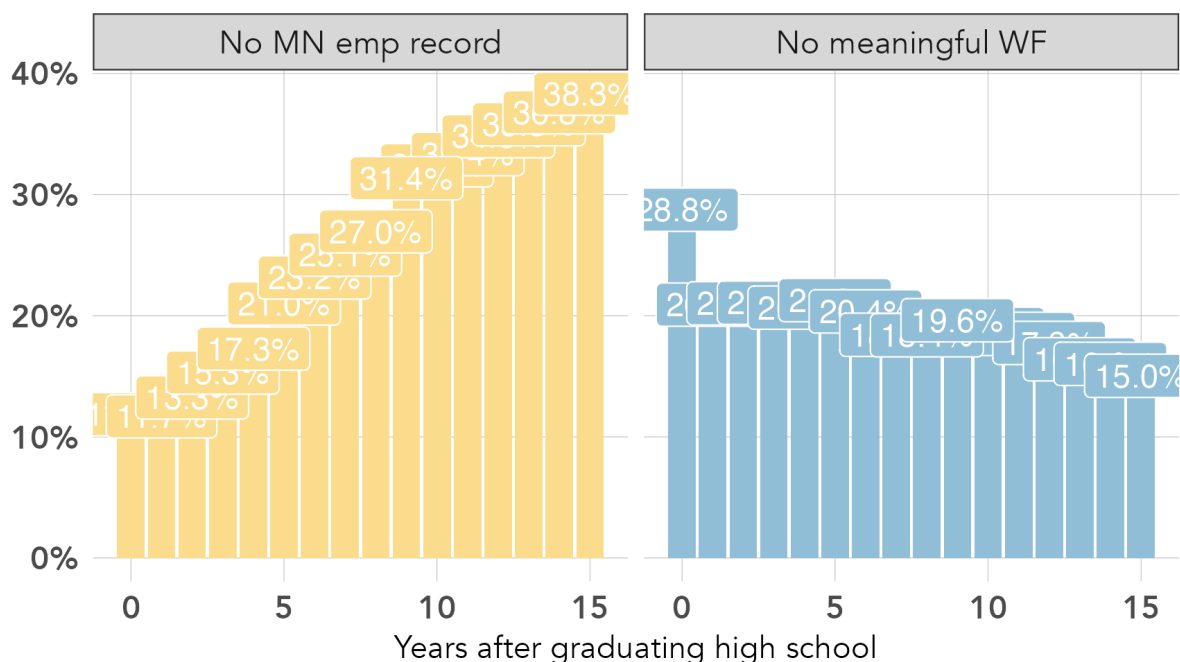


Chart X: Rates of “No MN Employment Record” and “Not Meaningful Employment” by Postsecondary Enrollment Status, 5 and 10 Years After Graduation.

In this sense, the Unanchored Pathway contains **two overlapping realities**:

- A **mobile segment**, employed successfully but outside Minnesota’s visibility; and
- A **disconnected segment**, struggling to establish stable work within the region.

Both reduce the local labor force, and compared with peers in the Anchoring or Academic pathways, those in the Unanchored group share several consistent characteristics:

- **Limited or no CTE engagement**, reducing early exposure to career options.
- **Lower academic proficiency** on the MCA exams.
- **Higher likelihood of economic hardship or special education participation.**

By five years after high school, the proportion of graduates in this category remains substantial—especially among non-college youth. Some reappear in Minnesota’s workforce records by the ten-year mark, but many do not.

From a regional perspective, this group represents more than missing data—it represents **missing connections**. Whether employed elsewhere or struggling locally, these graduates fall outside the systems designed to help them transition into stable adult roles.

3.4 Why These Pathways Matter

Together, these three pathways reflect the **diverse realities of rural graduates**.

- The **Anchoring Pathway** sustains local economies through applied skill and regional education awareness of opportunities.
- The **Academic Pathway** connects the region to broader markets and knowledge networks.
- The **Unanchored Pathway** reveals where systemic barriers persist — and where targeted policy can drive change.

The goal is not to merge all students into a single pipeline, but to **strengthen each pathway on its own terms**:

- Support and expand the Anchoring Pathway through CTE and employer partnerships.
- Stay connected to the Academic Pathway through alumni networks and return-to-region incentives.
- Rebuild the Unanchored Pathway through outreach, re-engagement, and adult training programs.

In short:

Northeast Minnesota’s graduates follow three main routes after high school. Each pathway tells a different story about opportunity — and together, they point the way toward a balanced strategy for regional prosperity.

4 - Anchoring pathways

The Anchoring Pipeline represents graduates who remain and build their lives in Northeast Minnesota. Their success depends on how strongly high school experiences connect to local postsecondary and employment systems.

4.1 Key Finding 1: CTE is the single strongest predictor of regional employment.

One of the most powerful predictors of whether a graduate gained meaningful workforce participation in **Northeast Minnesota** was their level of participation in **Career and Technical Education (CTE)**.

Because of how CTE programs are designed—developed in partnership with regional employers and aligned with local workforce needs—participation in CTE also serves as a **strong proxy for how connected students are to local and regional career opportunities.**

CTE programs are more than technical electives. They are, by definition, a *“sequence of courses that integrates core academic knowledge with technical and occupational skills, providing students a pathway to postsecondary education and careers.”*

The relationship between CTE engagement and regional employment was remarkably consistent and robust. Students who completed or concentrated in CTE programs—and even those who participated without completing a full sequence—were **10 to 15 percentage points more likely** to hold meaningful employment in Northeast Minnesota at one, five, and ten years after high school.

Five years after graduation, the difference is especially clear:

- **32%** of CTE concentrators/completers were working meaningfully within the region,
- **27%** of CTE participants (without full completion) achieved the same, while
- Graduates with **no CTE engagement** were significantly less likely to work in the region and more likely to appear in out-of-region categories—such as attending postsecondary institutions elsewhere, working outside the region, or having no Minnesota employment record.

Share of CTE Achievement by workforce state 5 years after graduating high school

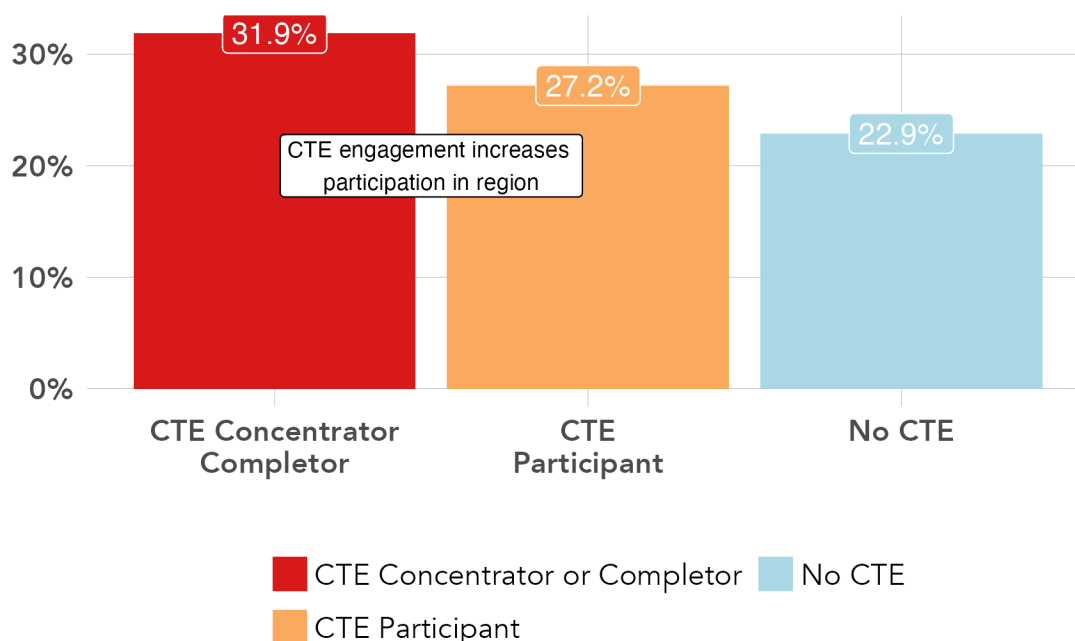


Chart X: As CTE engagement increases so does the likelihood of gaining meaningful workforce participation in the region. Data: MN Office of Higher Education - SLEDs

What makes this relationship particularly important is its **persistence**. Even when controlling for other factors—such as academic achievement, demographic characteristics, and economic background—CTE participation remained one of the strongest and most consistent predictors of local employment.

In other words, **students' exposure to hands-on, career-connected learning experiences can overcome many of the factors that typically pull young people away from their home region.**

Key takeaway:

CTE doesn't just prepare students for jobs—it connects them to place. By combining academic learning with regional relevance, CTE helps more graduates build meaningful careers and stable lives in Northeast Minnesota.

4.2 - Regional Higher Education Keeps Graduates Anchored

For graduates who continue their education after high school, **where** they go to college and **what** kind of credential they earn play a major role in whether they remain connected to Northeast Minnesota's workforce.

Three key factors stand out: **college location, college type, and degree level.**

The Power of Place: Graduating Locally, Working Locally

The data show a strong and consistent relationship between college location and workforce outcomes. Five years after high school, **half of graduates who completed a degree from a regional college** were working meaningfully within Northeast Minnesota — a significantly higher rate than graduates from colleges elsewhere in the state or outside Minnesota.

College institution location as percent of meaningful workforce participation 5 years after graduating high school

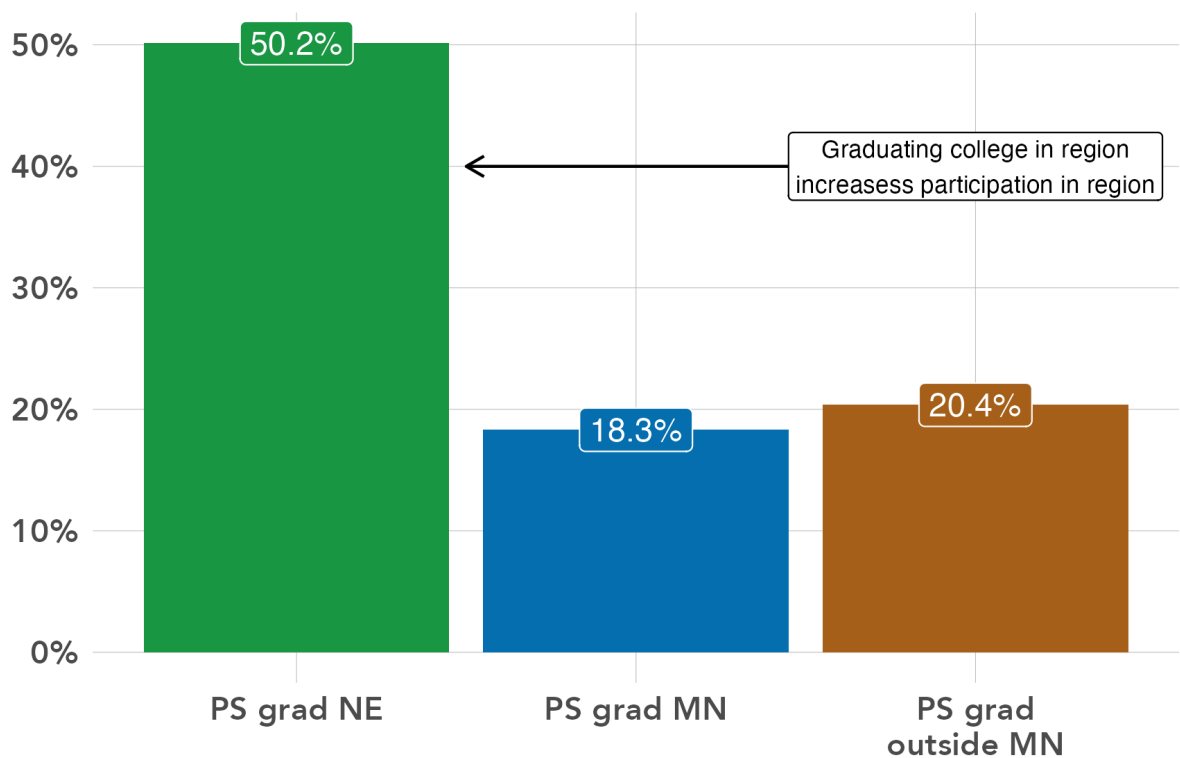


Chart X: Half of high school graduates in Northeast that complete a degree in a regional college have meaningful workforce participation in the region 5-years after high school. Data - MN Office of Higher Education - SLEDs

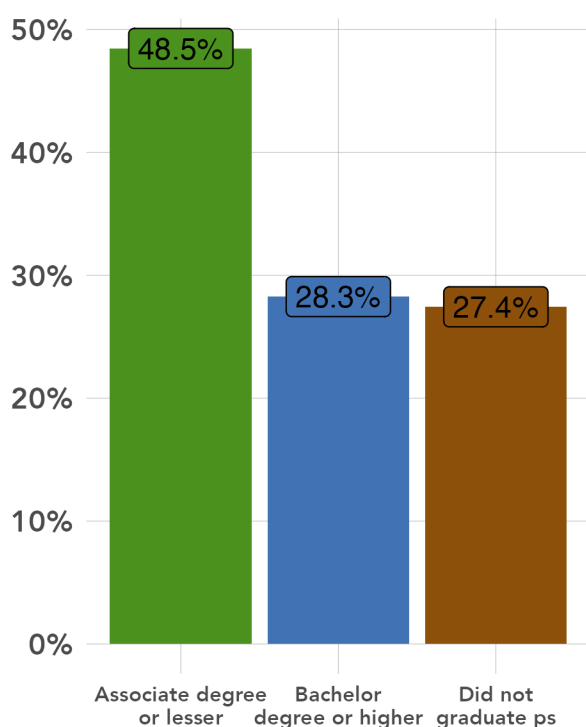
This finding underscores the importance of place-based higher education. Students who attend college locally often develop stronger networks with local employers, complete internships within the region, and are exposed to regional job opportunities while studying. These ties frequently translate into permanent employment after graduation.

Degree Type and College Type Also Matter

The connection between local education and local employment holds true across different credential levels and types of colleges. Roughly **half of Northeast graduates who earned an associate degree or lower credential** were meaningfully employed in the region five years after high school. The same was true for those who graduated from a **two-year public institution**, regardless of whether the college itself was located within the region.

Has meaningful workforce participation in the region 5-years after high school

Highest credential earned



Type of institution from which highest credential was earned

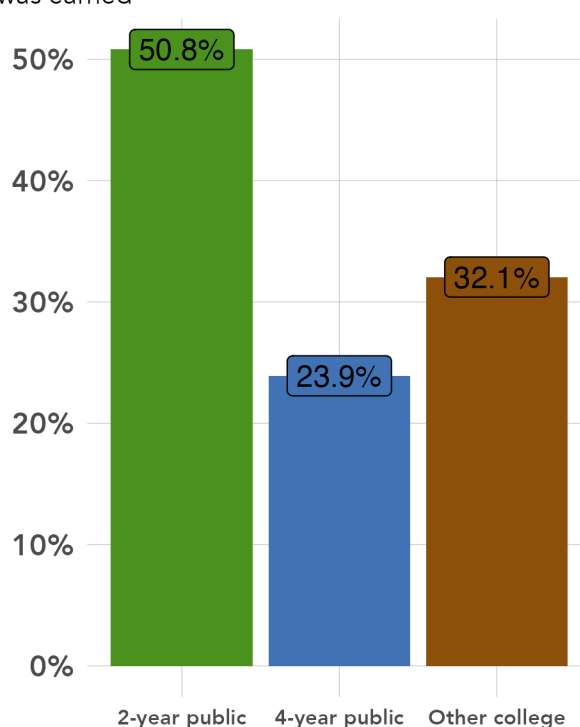


Chart X1: Half of the Northeast high school graduates that complete post-secondary from a 2-year college and/or earn an Associate Degree or lesser end up having meaningful workforce participation in the region. Data: MN Office of Higher Education - SLEDs

These outcomes highlight how **two-year colleges and applied credentials** serve as critical bridges between education and regional opportunity. Their programs are often designed around local workforce needs, making graduates naturally more employable in the region's core industries.

Encouraging Signs for the Regional Talent Pipeline

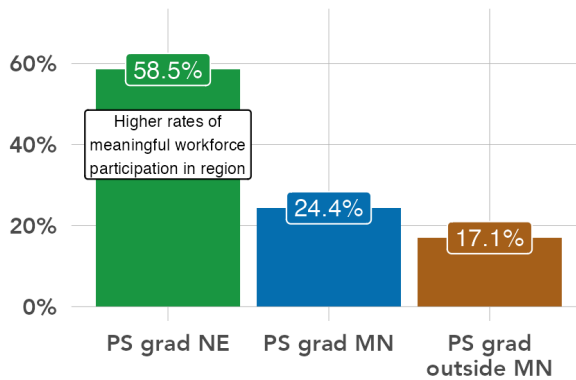
The best news for Northeast Minnesota is that **a large share of its college graduates already follow these anchoring pathways**. Among all high school graduates who went on to complete a college degree:

- **59%** graduated from a college located within the region,
- **43%** earned an **associate degree or lower**, and
- **39%** graduated from a **two-year public institution**.

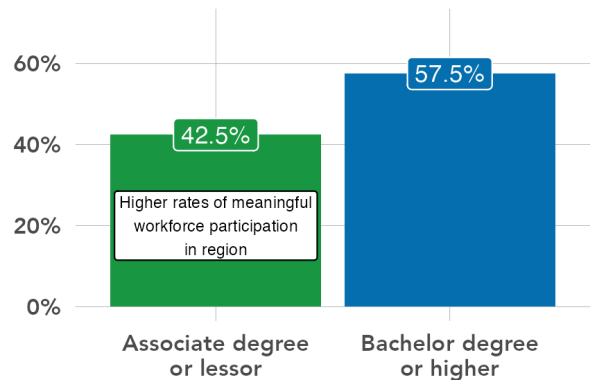
[What does this say about whether these students stayed in the region or not?]

Share of high school graduates that complete college

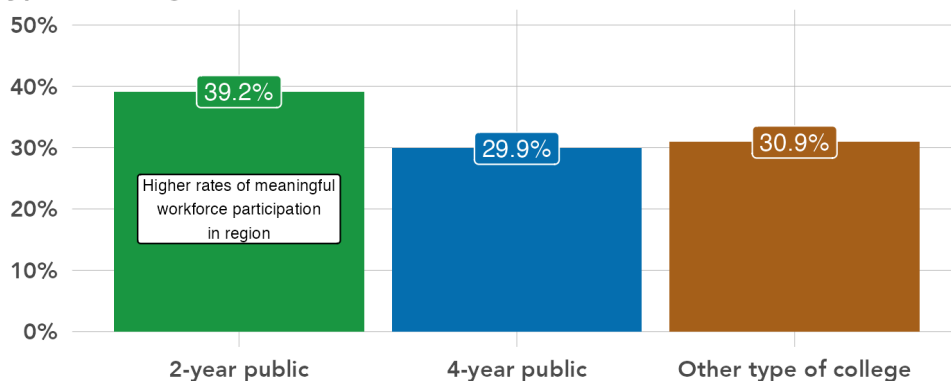
College location



Highest credential earned



Type of college



Each of these characteristics—regional college completion, two-year public institutions, and applied degrees—correlates strongly with higher rates of meaningful workforce participation in the region.

Even though **bachelor's degrees** remain the most common credential earned statewide (58% among Northeast graduates who completed college), the region's relatively high share of **associate degree holders and regional college alumni** is an advantage. These graduates are the most likely to stay, work, and contribute to the regional economy.

Key takeaway:

Higher education that happens *in* the region tends to stay *in* the region. Local colleges and two-year institutions don't just educate—they anchor graduates by connecting learning to local opportunity.

4.3 - Connecting Local Career Awareness and Higher Education Pathways

The relationship between **career-connected learning in high school** and **postsecondary choices after graduation** is one of the clearest findings in this study. Students who engage in **Career and Technical Education (CTE)** during high school are not only more likely to stay and work in the region — they are also more likely to choose **college pathways that keep them anchored** to the region's workforce.

In short, CTE doesn't just connect students to work; it connects them to **postsecondary pathways that lead back home**.

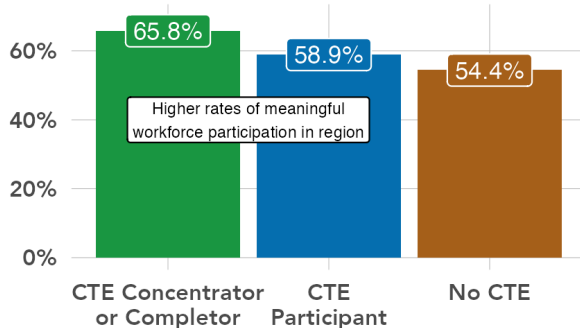
CTE as a Bridge Between High School and Local College Pathways

Graduates who were **CTE concentrators, completers, or even moderate participants** showed significantly higher rates of:

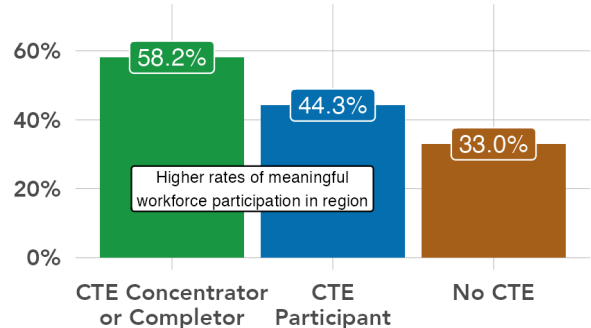
- Attending **colleges located within the region**,
- Enrolling in **two-year public institutions**, and
- Earning **associate degrees or lower credentials**—the very pathways most strongly associated with meaningful workforce participation in Northeast Minnesota.

Share of CTE achievers

Graduated from a NE college



Earned and Associate Degree or lesser



Graduated from a 2-year college

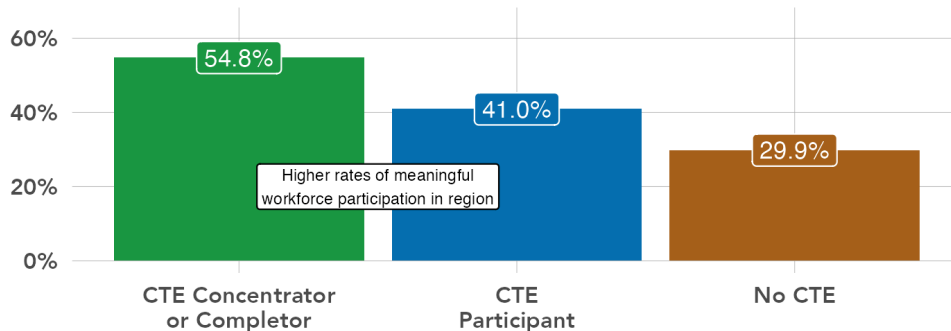


Chart X: Higher percentages of individuals that are more heavily involved with CTE coursework tend to follow post-secondary paths that lead to meaningful workforce participation in the region. Data: MN Office of Higher Education - SLEDs

These relationships are more than statistical—they reflect how CTE programs are built. Because CTE curricula are developed in collaboration with local employers and regional colleges, students in these programs naturally gain early awareness of **career options that exist close to home**.

That awareness, coupled with relevant hands-on experience, makes it far more likely that CTE students will **see local higher education as a logical next step**, not a fallback option.

This alignment between CTE and postsecondary pathways creates what might be called a **regional retention loop**:

1. CTE engagement introduces students to local industries and regional postsecondary programs.
2. Students pursue applied education at regional or two-year colleges.
3. These institutions prepare them for occupations that exist locally.

4. Graduates find meaningful work in the region's core industries.

It's a virtuous cycle that strengthens the regional labor force from within.

But this pattern also highlights the challenge on the other side of the equation. If students lack early exposure to local career options—through CTE, work-based learning, or employer partnerships—they are **less likely to see the region's colleges or industries as viable pathways**, increasing the odds they'll leave for education or work elsewhere.

Key takeaway:

CTE doesn't just prepare students for immediate employment — it shapes their educational trajectories. By connecting classroom learning to local industries and regional colleges, CTE builds a bridge between high school and higher education that keeps more graduates rooted in Northeast Minnesota.

5 - Mobility Pathways

The **Mobility Pathways** illustrate how education can expand opportunity beyond Northeast Minnesota. These routes often reflect success—graduates who pursue higher education and careers that take them elsewhere—but they also signal how academic and institutional factors shape geographic outcomes.

Two main patterns emerge:

1. **Meaningful Workforce Participation in Minnesota but Outside the Region**, and
2. **No Minnesota Employment Record**, representing graduates whose careers or education extend beyond state data coverage.

Both pathways involve **leaving the region**, but for different reasons and in different contexts.

5.1 - Meaningful Workforce Participation Outside Northeast Minnesota

For many graduates, mobility within Minnesota reflects a logical next step in their academic and career trajectories. The data show that **earning a four-year degree and attending college outside the region** are the strongest predictors of meaningful employment elsewhere in the state. In short, both **college location** and **credential level** drive mobility. Graduates who pursue advanced degrees outside the region are much more likely to find career opportunities elsewhere in Minnesota's labor market.

Five years after graduation, **44% of students who completed college outside Northeast Minnesota but within the state** had meaningful employment elsewhere in Minnesota. In contrast, fewer than **20%** of graduates in other categories achieved the same outcome.

Share of college graduates in Minnesota but outside the region by workforce participation categories 5-years after high school

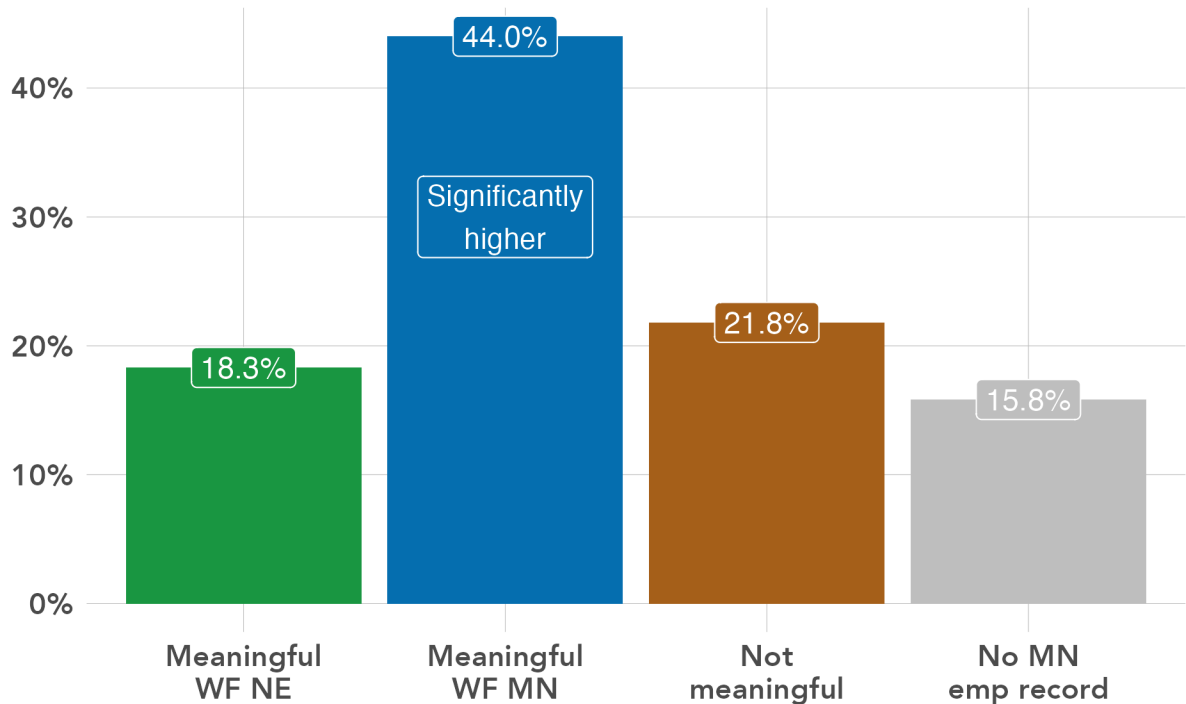


Chart X: Individuals who graduated from a college outside the region but within the state tend to have higher rates of meaningful workforce participation in Minnesota but outside Northeast. Data: MN Office of Higher Education - SLEDs

The type of college also matters. Graduates from **four-year public institutions**—regardless of location—were significantly more likely to work meaningfully in Minnesota but outside the region (**32%**) than in other workforce categories (roughly 20%).

Earning a **bachelor’s degree** shows a similar pattern: **30%** of bachelor’s degree holders worked meaningfully outside Northeast Minnesota, compared to **28%** within it.

Share of 4-year college grads and Bachelor degree earners by workforce participation category

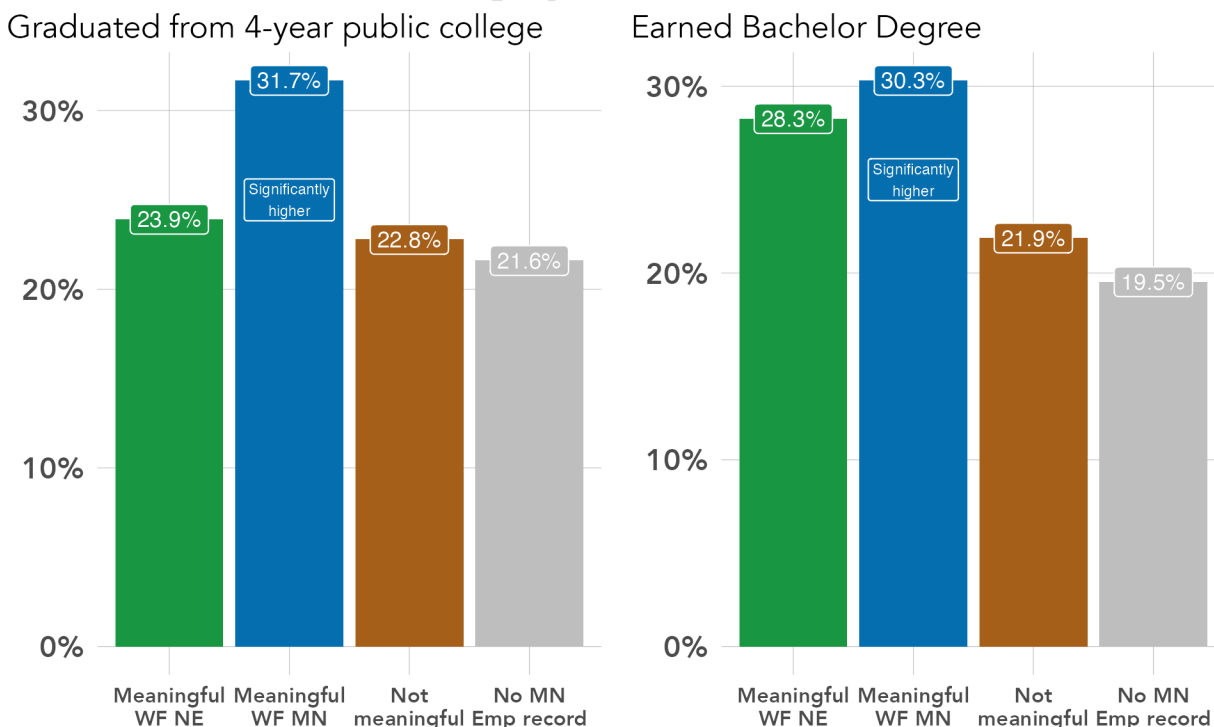


Chart X: The percentage of individuals that graduate from a 4-year college institution has a significantly higher likelihood of having meaningful workforce participation 5-years after high school. Data: MN Office of Higher Education - SLEDs

In short, both **college location** and **credential level** drive mobility. Graduates who pursue advanced degrees outside the region are much more likely to find career opportunities elsewhere in Minnesota's labor market.

The Academic Drivers of Mobility

So, what leads some students to follow these outward pathways? The analysis consistently points to a familiar set of factors: **academic preparation, college readiness activities, and limited connection to local career programming** during high school.

Students who excel in traditional academic measures—**higher MCA Math, Reading, and Science scores, ACT participation, and Post-Secondary Enrollment Options (PSEO)**—were significantly more likely to:

- Attend college outside the region, and
- Later achieve meaningful employment elsewhere in Minnesota.

For instance, **24% of students who exceeded proficiency in MCA Math** held meaningful jobs in Minnesota but outside Northeast five years after graduation—*nearly double* the rate for students who were below proficiency (**12.6%**).

Share of MCA-Math proficiency scores in meaningful workforce participation outside NE 5-years after high school

Higher MCA-Math scores increases the likelihood of having meaningful workforce participation outside the region

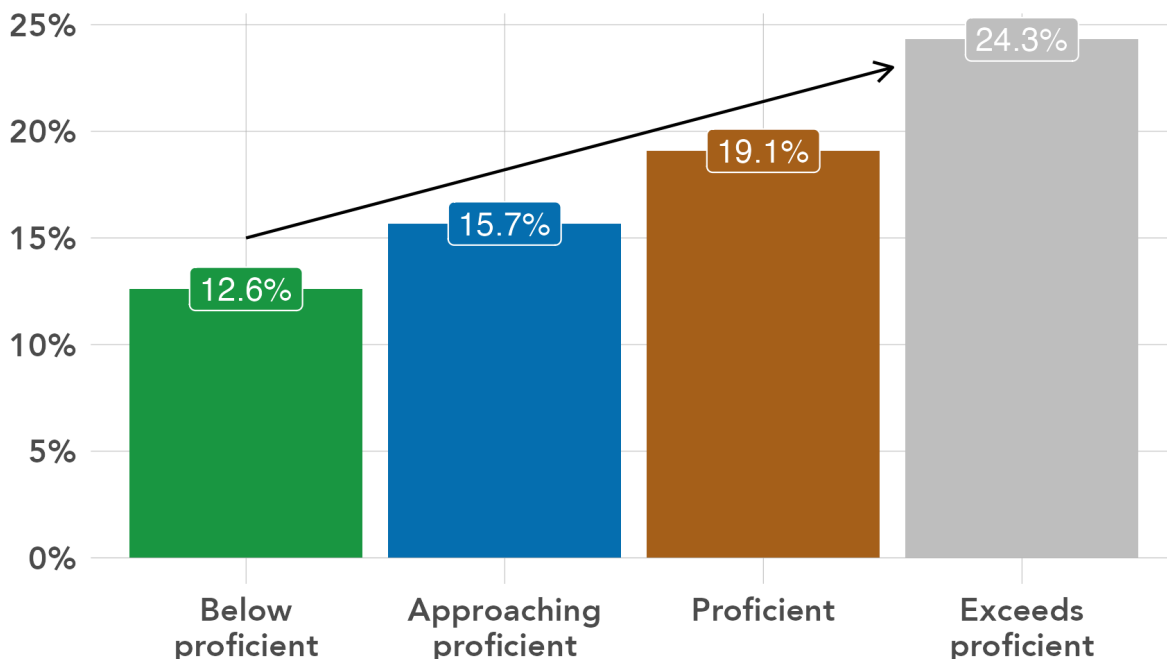


Chart X: As MCA - Math proficiency increases, the likelihood of having meaningful workforce participation increases as well. Data: MN Office of Higher Education - SLEDs

This pattern extends across other measures of academic readiness. Students who took the ACT, earned higher composite scores, or participated in PSEO were **10 or more percentage points more likely** to work outside the region than those who did not.

[insert some sort of graphic highlighting these relationships - no numbers]

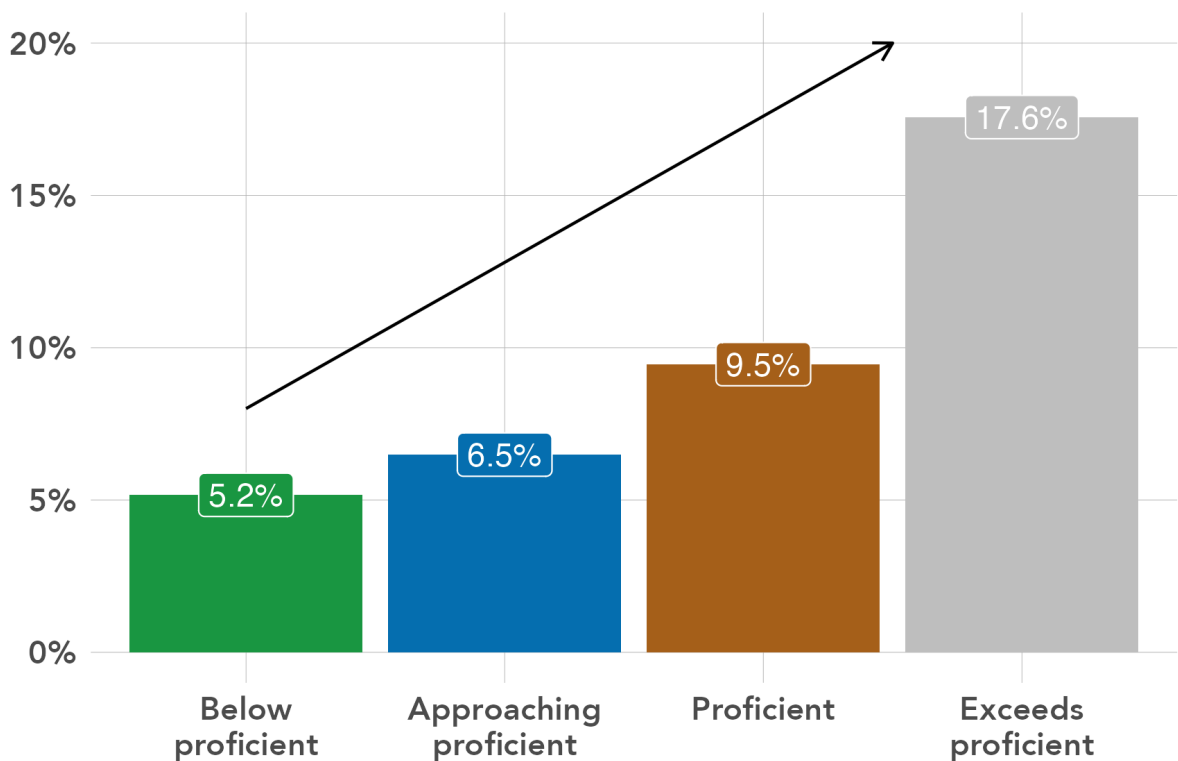
These findings show that high academic achievement doesn't directly cause mobility—it **enables** it. Students with strong academic credentials gain access to a broader range of postsecondary options, which in turn increases their likelihood of studying and eventually working outside the region.

From Academic Readiness to Postsecondary Mobility

The link between academic preparation and geographic mobility becomes even clearer when examining where students attend college.

- Among students who **exceeded MCA Math proficiency**, **18% graduated from a college outside Northeast Minnesota**, compared to only **5%** of those below proficiency.
- Similarly, students who took the **ACT**, achieved high composite scores, or participated in **PSEO** were **three to five times more likely** to graduate from a college outside the region.

Share of MCA-Math proficiency scores by graduating from a college located in MN but outside the region



*Chart X: As math scores increase so does the likelihood that they will graduate from a college outside Northeast.
Data: MN Office of Higher Education - SLEDs*

Proportion that graduated from a college outside the region

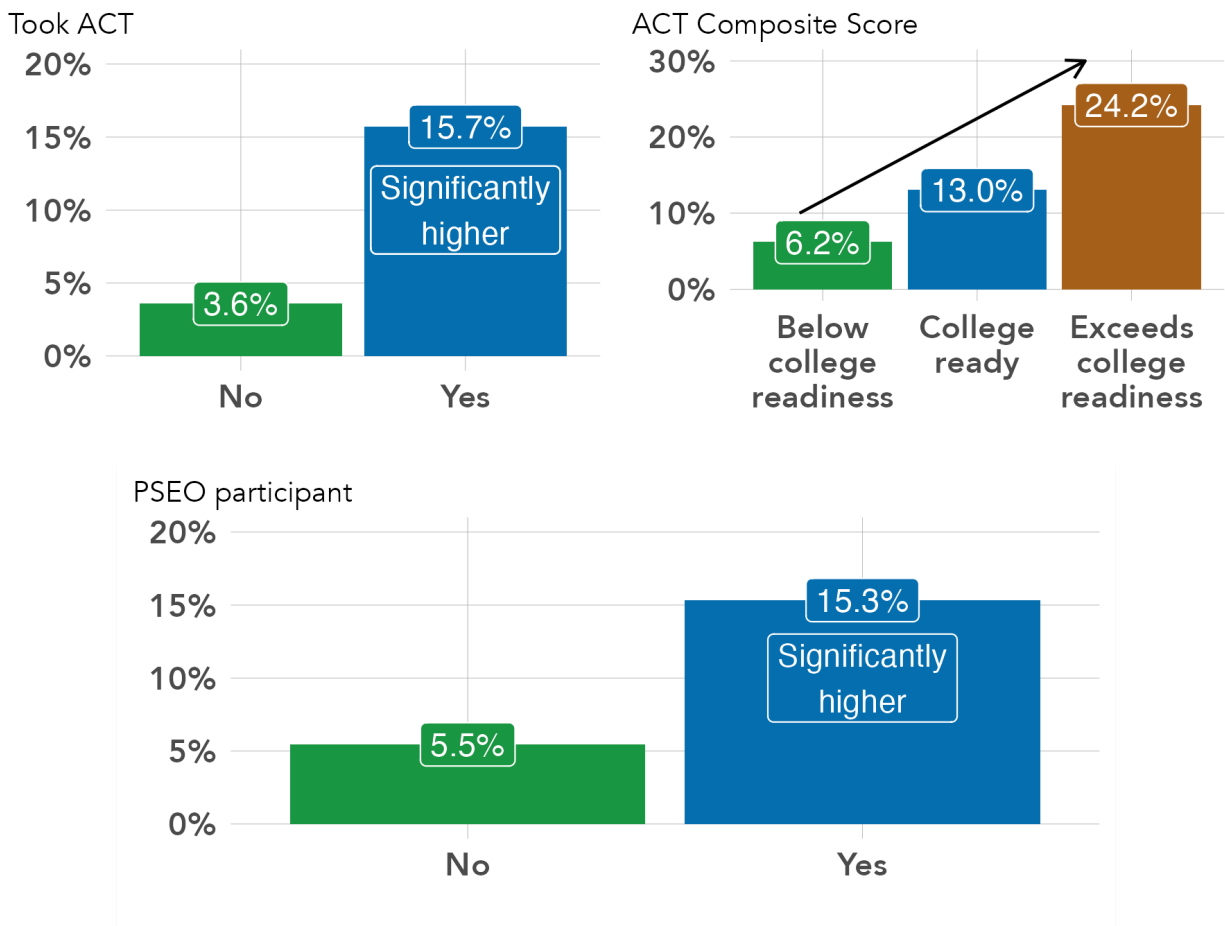


Chart X: College assessment tests and engaging in post-secondary opportunities significantly increases the likelihood that a high school graduate in Northeast will graduate from a college outside Northeast. Data: MN Office of Higher Education - SLEDs

Together, these results paint a consistent picture:

Academic readiness opens doors—geographically as well as educationally.

High-performing students are more likely to pursue four-year institutions outside Northeast Minnesota, leading to higher degrees and expanded career mobility across the state.

This mobility is not inherently negative. These graduates are succeeding—earning degrees, finding stable work, and contributing to Minnesota’s economy. But their patterns also underscore the **tension between opportunity and regional retention**: the same factors that create success for individuals can, collectively, draw talent away from the region.

Key takeaway:

Academic preparation and college readiness activities—ACT testing, higher MCA scores, and PSEO participation—don’t directly push students out of the region, but they **open pathways that make mobility more likely**. For policymakers, this reinforces the importance of balancing academic rigor with strong regional connections, ensuring that opportunity doesn’t mean permanent outmigration.

5.2 - No Minnesota Employment Record and Postsecondary Pathways

The category “**No Minnesota Employment Record**” captures a complex mix of outcomes. Some graduates in this group may be **working successfully outside Minnesota**, while others are **serving in the military, employed by the federal government, self-employed, or temporarily out of the labor force**. It also includes a smaller number of individuals who remain in the state but are **unemployed or in informal work** not captured in state wage data.

In other words, “*No Minnesota employment record*” does not necessarily mean disconnection—it often reflects **mobility that the data can’t fully see**. Yet the underlying patterns show that this group behaves similarly to the **mobile pathways** than the disconnected ones.

Out-of-State College Graduates and Workforce Mobility

The clearest evidence of this comes from college location. Graduates who complete college **outside Minnesota** are far more likely to fall into the “No MN employment record” category.

Five years after high school, **40% of graduates from out-of-state colleges** had no Minnesota employment record—roughly **double** the rate of other workforce participation categories, which hover around 20%.

Share that graduated from a college outside Minnesota by workforce category 5-years after high school

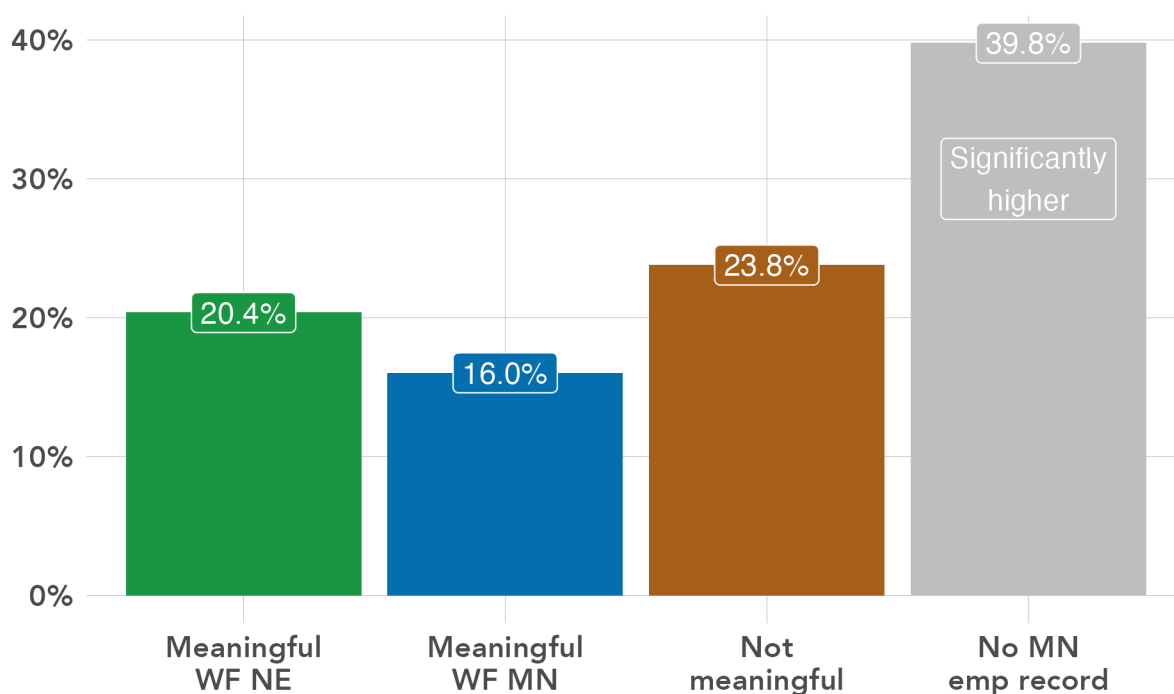


Chart X: Graduating from a college outside Minnesota greatly increases the likelihood of having no Minnesota employment record. Data: MN Office of Higher Education - SLEDs

This relationship mirrors the pattern seen with in-state mobility. Just as graduates of Minnesota colleges located outside Northeast Minnesota are more likely to work meaningfully elsewhere in the state, graduates of colleges **outside Minnesota entirely** are more likely to find meaningful employment **beyond state borders**.

While the data cannot capture their exact destinations or career outcomes, the consistency of this pattern suggests that many of these individuals are **successfully employed elsewhere**, not disengaged.

High School Preparation and Out-of-State Postsecondary Pathways

The factors that lead students to attend college outside Minnesota are strikingly similar to those that drive in-state mobility. High academic achievers—those who take the **ACT**, earn higher **composite scores**, and participate in **Postsecondary Enrollment Options (PSEO)**—are significantly more likely to pursue college outside the state and, by extension, to appear in the “No MN employment record” category.

For example, students with high ACT scores or PSEO participation were several times more likely to complete college outside Minnesota than peers who did not. This again reinforces that **academic strength and exposure to postsecondary opportunity** often translate into **geographic mobility**, not necessarily workforce disengagement.

Proportion that graduated from a college outside Minnesota

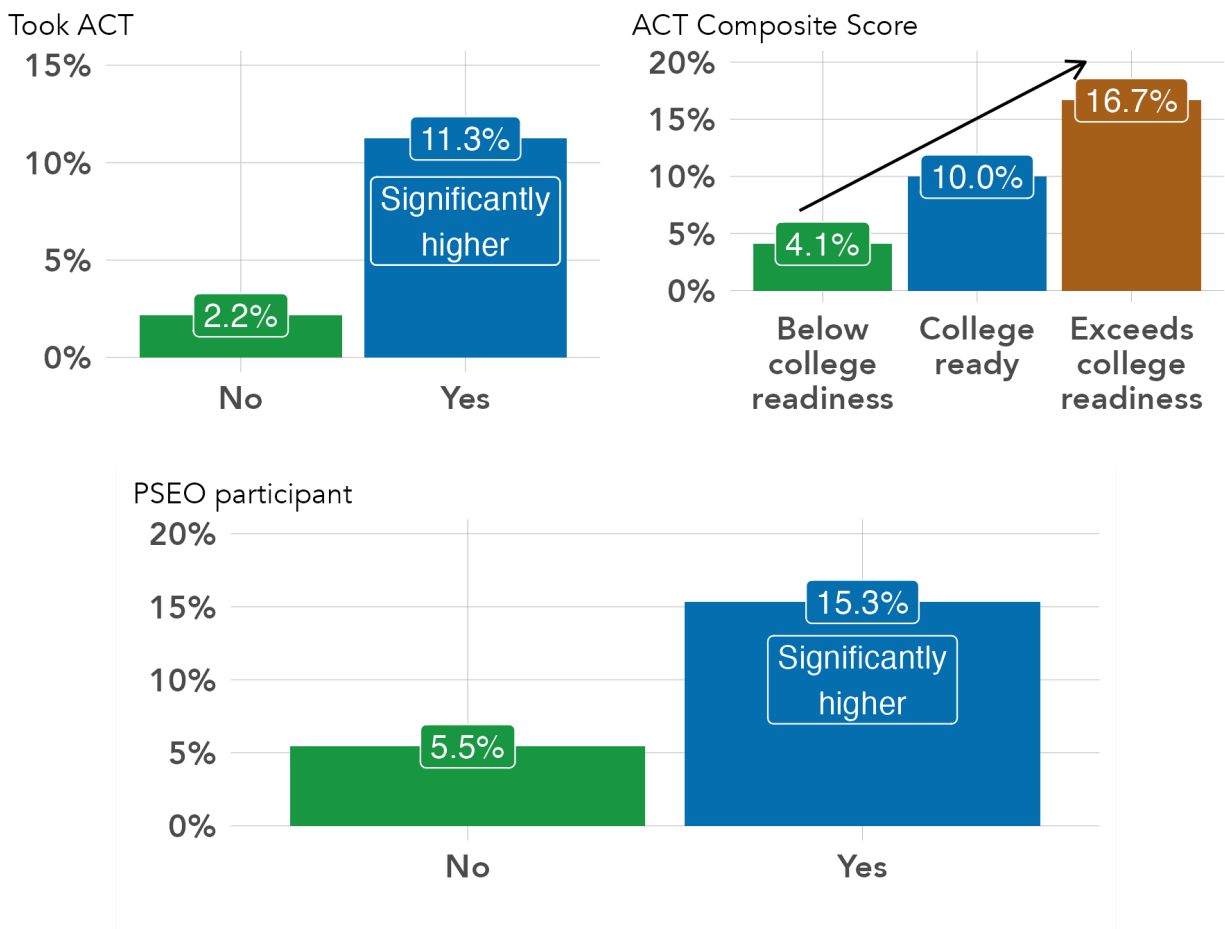


Chart X: Higher ACT scores and PSEO participants have significantly higher chances of completing college outside the state, and likely having meaningful workforce participation in that state. Data: MN Office of Higher Education - SLEDs

Balancing Opportunity and Retention

These findings highlight one of the most difficult—and important—questions facing regions like Northeast Minnesota:

*How can communities **celebrate mobility and ambition** while also **retaining the talent they help develop**?*

It's not realistic—or desirable—to discourage young people from pursuing out-of-state opportunities. Mobility can represent success: students expanding their horizons, gaining new experiences, and building careers that may one day bring them back home.

However, regional leaders must also ask whether youth are **leaving by choice or by default**. Are they drawn away because they see greater opportunity elsewhere, or because they lack awareness of strong local options in education and careers?

If students make informed choices—knowing the full range of opportunities in the region—then outward mobility is a healthy sign of success. But if they leave based on incomplete information or misconceptions about the region's economy, then there's room for intervention.

The story of mobility in Northeast Minnesota is largely one of opportunity — graduates pursuing education and careers that take them beyond the region and, in many cases, beyond the state. These young people represent success in a broader sense: they are working, learning, and contributing to economies across Minnesota and the nation.

But while some graduates leave by choice and with clear purpose, others fall away from both education and stable work for different reasons. Their stories are not about mobility, but about **disconnection**. They are the students who never fully transition into postsecondary programs or secure meaningful employment, and for whom the bridge between school and work simply never solidifies.

This next section focuses on that group — the **Unanchored Pathway**. Unlike the mobile graduates whose opportunities lie elsewhere, these individuals remain largely unseen in the data and underserved by the systems meant to support them. Understanding their patterns is essential, because in many ways, **they represent the greatest potential — and the greatest challenge — for building a stronger regional workforce.**

6 - Falling through the cracks

The previous sections showed how academic preparation, college attendance, and Career and Technical Education (CTE) participation influence whether graduates build meaningful careers in or beyond the region. But what about the young people who don't follow those paths — those who don't finish college and never quite gain a foothold in stable work?

This section focuses on that group: the individuals most at risk of **falling through the cracks** of the education-to-employment system. Their outcomes highlight both the barriers that persist and the opportunities available to help more graduates find meaningful, lasting participation in the regional workforce.

6.1 Non-College Graduates: The Group Most at Risk

The data make one thing clear: **not completing college is the single biggest predictor of disconnection from meaningful work.**

A majority of Northeast Minnesota's high school graduates — about **58%** — do not complete college. This is a large share of the population and, as Chart X shows, their postsecondary outcomes diverge sharply from those of college graduates.

Five years after high school, nearly **six in ten non-college graduates (59%)** are either:

- employed in Minnesota or the region but **not meeting the threshold for meaningful employment**, or
- have **no Minnesota employment record at all.**

Percent of non-college grads by workforce participation category

Nearly 60% of non-college grads end up without meaningful workforce participation in Minnesota

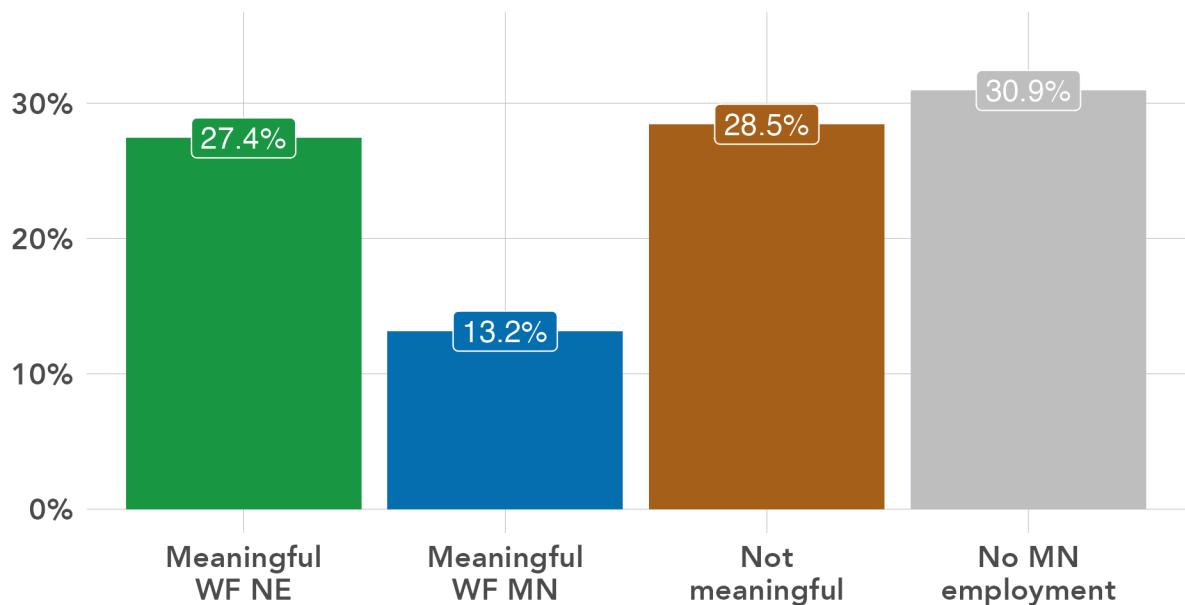


Chart X: A majority of non-college graduates fall into either not having meaningful workforce participation or having no Minnesota employment. Data: MN Office of Higher Education - SLEDs

Yet within this broader group lies a bright spot: roughly **27%** of non-college graduates do achieve meaningful workforce participation **in the region**. This share is **twice as high** as those who find meaningful employment outside the region (13%). These individuals demonstrate that

local opportunity exists — but that reaching it often depends on specific experiences and supports.

6.2 What Anchors Non-College Graduates Locally?

Among non-college graduates, the factor most strongly tied to meaningful workforce participation in the region is **engagement in CTE**.

Graduates who completed or concentrated in CTE were **significantly more likely** to work meaningfully in the region, while those with little or no CTE exposure were far more likely to appear with no Minnesota employment record. This finding reinforces what earlier sections demonstrated: CTE creates a direct line between education and work, even for those who don't pursue or complete a postsecondary degree.

Proportion of CTE achievement by workforce participation categories - Non-college grads

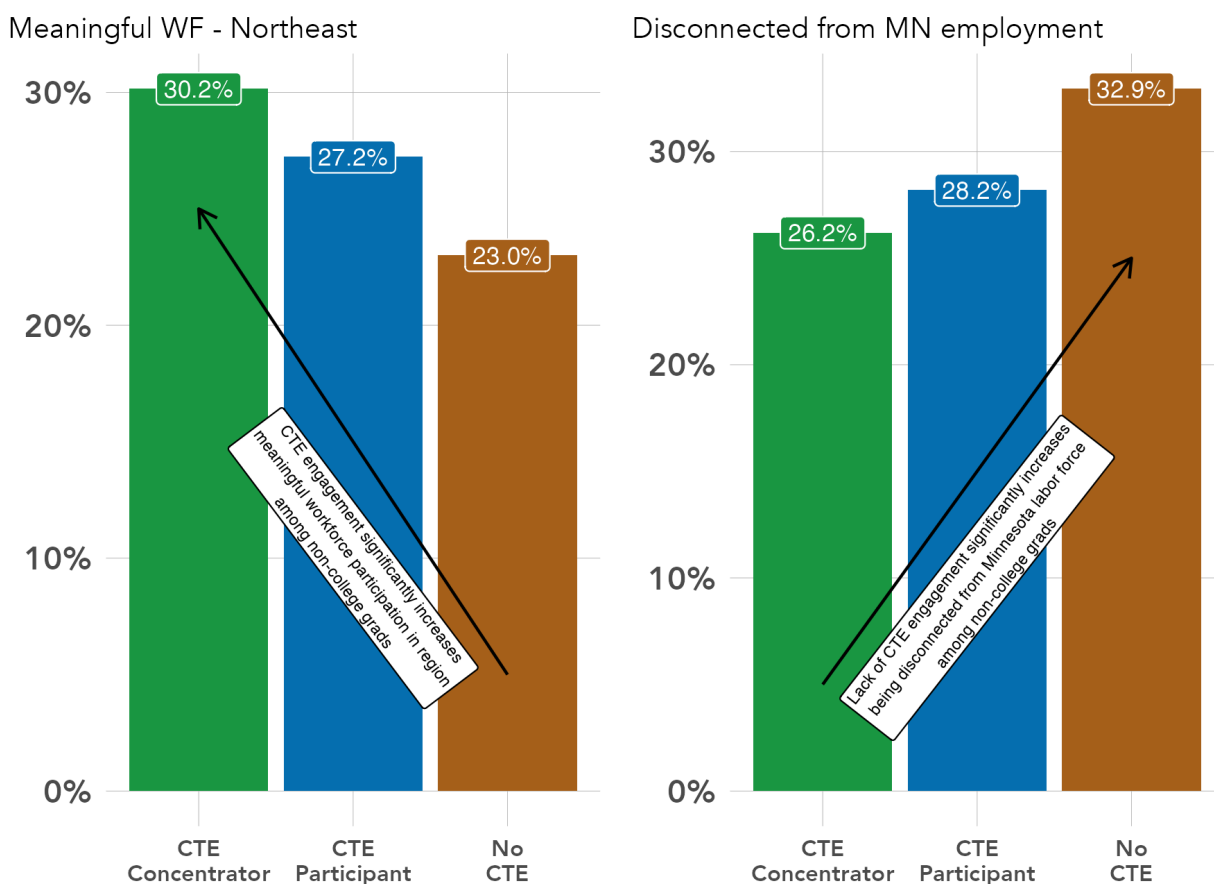


Chart X: CTE engagement increases the likelihood of having meaningful workforce participation in the region and

less likely to be disconnected from the Minnesota workforce among non-college grads. Data: MN Office of Higher Education - CTE

6.3 The Role of Socioeconomic Barriers

Beyond education, **socioeconomic factors** play a major role in shaping who becomes disconnected from meaningful work. Lower-income students and those who received special education services face barriers that begin in high school and extend into adulthood.

Income Status

Students who qualified for free or reduced-price lunch — a proxy for household income — were **three times more likely** not to graduate from college (**74% vs. 26%**).

Free or reduced lunch eligibility by college completion

Lower income students significantly more likely to not graduate from college

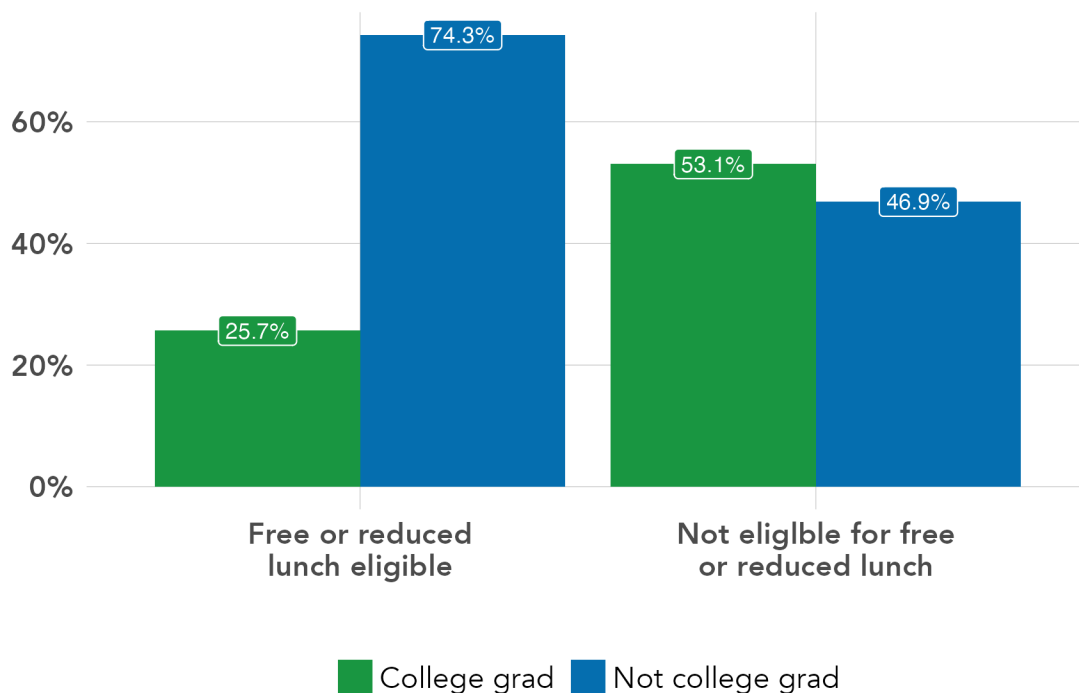


Chart X: Individuals that come from lower income households are three times more likely to not graduate college. Data: MN Office of Higher Education - SLEDS

This matters because college completion is one of the strongest drivers of upward mobility for students from lower-income households. Among low-income graduates who completed college, more than **60% achieved meaningful workforce participation within Minnesota**, with **41%**

working meaningfully in Northeast Minnesota and another **21%** employed meaningfully elsewhere in the state. Only **15%** of this group had no Minnesota employment record, and fewer than **23%** were in jobs classified as “not meaningful.”

By contrast, outcomes for low-income students who did **not** complete college look very different. Only **38%** reached meaningful workforce participation of any kind—**25% in the region** and just **12% elsewhere in the state**. Meanwhile, over **62%** of non-college, low-income graduates either lacked a Minnesota employment record or were working in jobs that did not meet the meaningful participation threshold.

Lower income students - college and non-college grads by workforce participation category

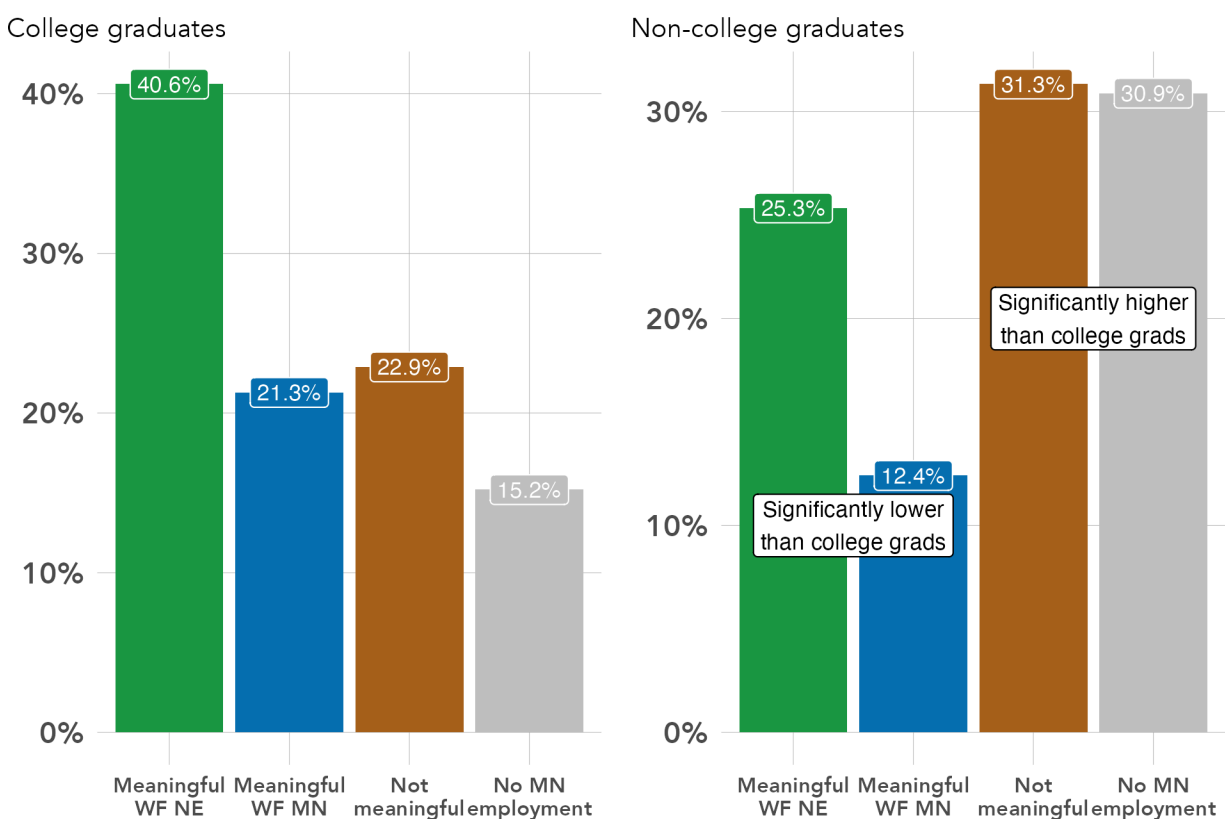


Chart X: Lower income individuals that graduate from a college are significantly more likely to have meaningful workforce participation compared to their peers that do not graduate from college. Data: MN Office of Higher Education - SLEDs

Taken together, these figures show that **college completion more than doubles the likelihood of meaningful workforce participation for low-income graduates**. It not only opens the door to higher-wage and stable employment, but also appears to strengthen the likelihood of remaining engaged in the state’s workforce—particularly within Northeast Minnesota’s regional economy.

Special Education

A similar pattern appears among students who received **special education services** in high school. While these graduates face additional barriers to completing college, the data show that earning a postsecondary credential dramatically improves their connection to the workforce.

Among special education students who **completed college**, nearly **60% achieved meaningful employment within Minnesota**, including **45% working meaningfully in Northeast Minnesota** and another **15%** employed meaningfully elsewhere in the state. Only **13%** had no Minnesota employment record, and about **27%** were employed in positions not meeting the “meaningful workforce participation” threshold.

By comparison, the picture is much more concerning for special education students who **did not complete college**. Only about **32%** reached meaningful workforce participation of any kind — **23% in the region** and **9% elsewhere in the state** — while more than **67%** had no Minnesota employment record or were working in non-meaningful jobs.

Received special education - college and non-college grads by workforce participation category

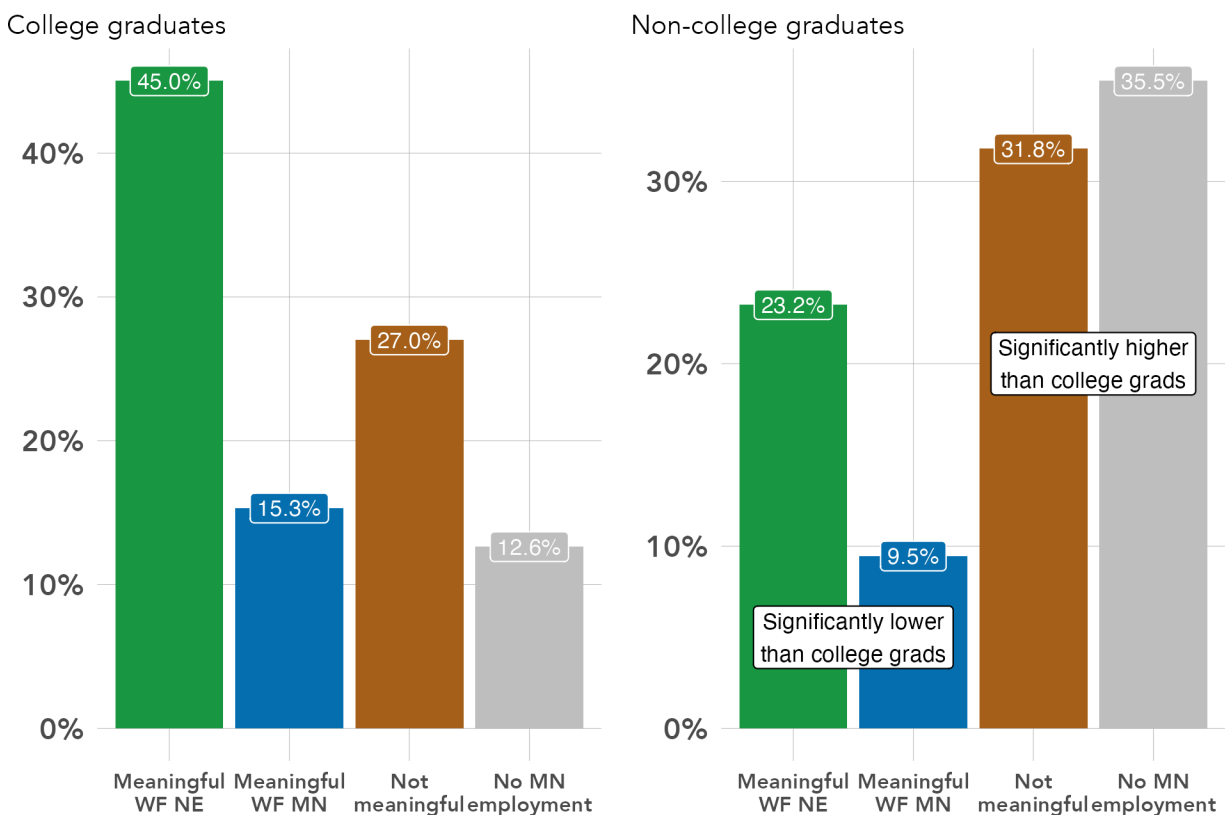


Chart X: Graduating from college can help break down barriers to meaningful workforce participation among individuals that received special education in high school. Data: MN Office of Higher Education - SLEDs

These findings highlight that **college completion substantially narrows the employment gap** for students with disabilities. Even though special education students remain underrepresented among college graduates, those who do earn a degree show outcomes nearly on par with their peers without special education backgrounds. The takeaway is clear: **when given equitable access to postsecondary success, special education students succeed — and contribute meaningfully to the regional economy.**

6.4 CTE as a Bridge for At-Risk Youth

Despite these challenges, there is a consistent pathway that improves outcomes for at-risk students: **Career and Technical Education.**

CTE doesn't erase inequality, but it helps **narrow the gap.** Among non-college graduates from low-income households, those who participated in CTE were **four percentage points more likely** to achieve meaningful workforce participation in the region.

Non-college grads eligible for free or reduced lunch with meaningful workforce participation in the region by CTE engagement

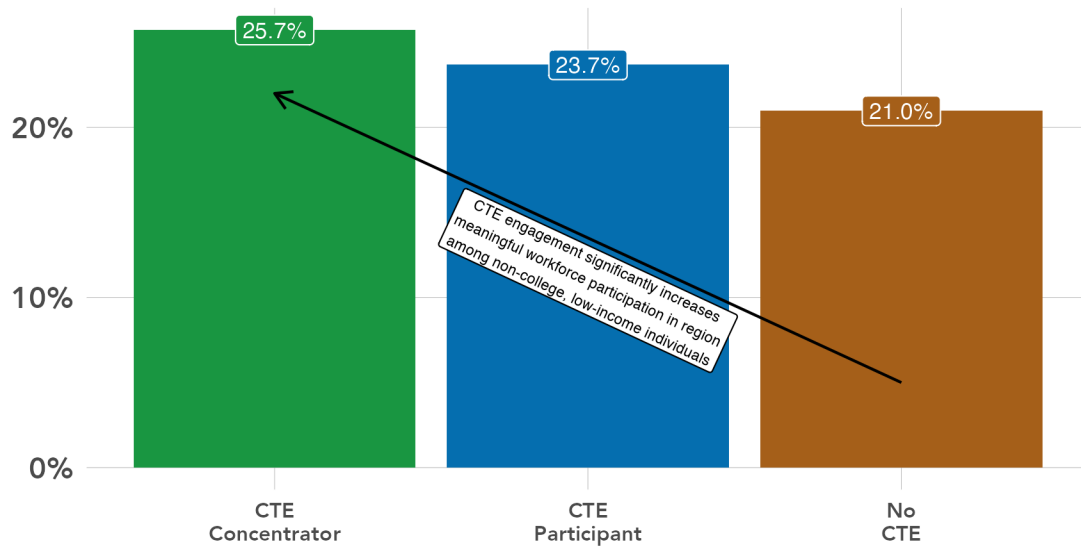


Chart X: Among non-college grads that qualify for free or reduced lunch, engagement in CTE increases the likelihood that they will have meaningful workforce participation in the region by 4 percentage points.

The same holds true for students who received special education services. Those who were CTE concentrators or completers were **significantly more likely** to work meaningfully in the region compared to peers with little or no CTE engagement.

Non-college grads requiring special education with meaningful workforce participation in the region by CTE engagement

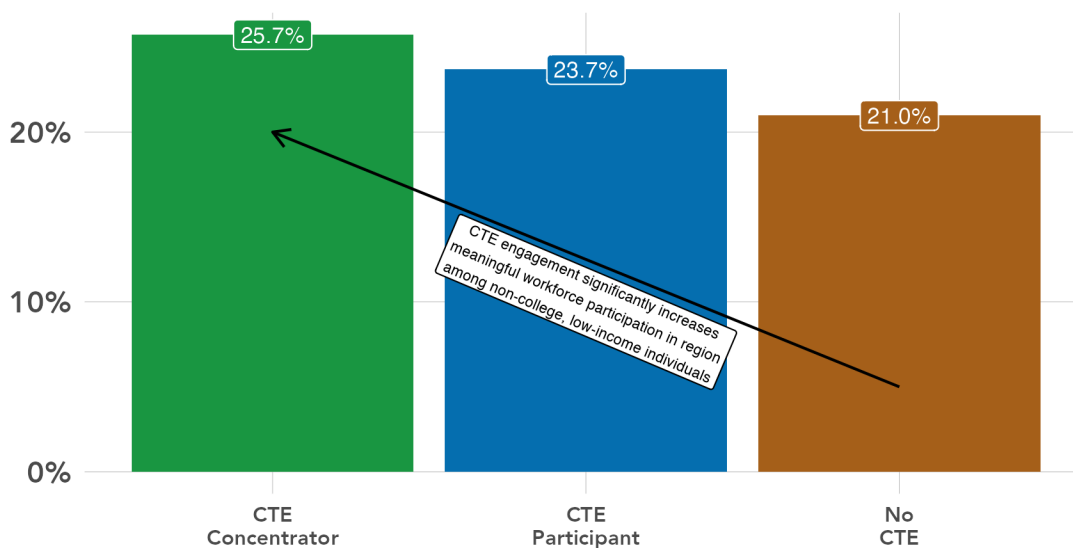


Chart X: Individuals that didn't graduate from college and required special education in high school and were CTE concentrators / completors were significantly more likely to have meaningful workforce participation in the region.
Data: MN Office of Higher Education - SLEDs

These findings point to CTE as one of the few interventions that consistently improves outcomes **across demographic lines**. It gives students—especially those least likely to complete college—a concrete connection to work, confidence in their skills, and a clearer vision for their place in the regional economy.

6.5 Looking Ahead: Turning Vulnerability into Opportunity

The evidence suggests that **non-college graduates are not disconnected by choice—they are disconnected by circumstance**. They face economic and educational hurdles that make postsecondary success harder to reach and workforce entry more fragile.

But the data also show that these challenges are **not insurmountable**.

CTE engagement, work-based learning, and employer partnerships can help translate potential into participation—especially for students from lower-income families or those receiving special education services.

For policymakers and educators, this points to a clear regional strategy:

Strengthen and expand CTE not only as a workforce development tool, but as a form of equity—one that connects students who might otherwise fall

through the cracks to the skills, confidence, and community networks that lead to meaningful work.

Taken together, the findings across all three pathways tell a clear story about opportunity and choice in Northeast Minnesota. The region's graduates are not following a single route after high school — they are navigating a landscape shaped by education, geography, and access to connection. Some are **anchored locally** through strong CTE programs and regional colleges. Others pursue **academic mobility**, earning degrees and building careers elsewhere in the state or beyond. And still others — the **unanchored or at-risk** — struggle to find their footing in work or education, often because the systems designed to support them don't yet reach far enough.

Each of these patterns carries lessons for regional policy. The Anchoring Pathway shows the power of local connection; the Mobility Pathway highlights the importance of maintaining ties with graduates who leave; and the Unanchored Pathway reveals the cost of disconnection — and the potential for transformation through better alignment, awareness, and support.

What comes next is about **action**. The following section outlines concrete strategies to strengthen all three pathways — helping the region not only retain more of its young talent, but also ensure that every graduate, wherever they go, remains part of the region's story.

7. Recommendations and Outlook

The findings from this study highlight an urgent but hopeful truth: the pathways that young people follow after high school are not fixed. They are shaped by experiences, opportunities, and the systems that connect education to work. Strengthening those connections — from high school through college and into meaningful employment — is the key to building a stronger, more inclusive regional workforce.

The following recommendations outline how Northeast Minnesota and the state can act on these insights.

1. Expand and Modernize Career and Technical Education (CTE)

Across every analysis, CTE emerged as one of the most powerful predictors of meaningful workforce participation in the region. CTE serves both as a workforce development strategy and as an equity intervention — helping students from diverse backgrounds, including those less likely to complete college, connect to meaningful local work.

Action steps:

- **Invest in modern CTE infrastructure:** update labs, equipment, and digital tools to match current industry standards.
- **Support teacher recruitment and professional development** to sustain high-quality programming in rural districts.
- **Integrate CTE with local employers and regional colleges,** creating clear “learn and earn” pathways through internships, apprenticeships, and dual-credit programs.

Policy impact: Modern, high-quality CTE ensures that more students — especially non-college graduates — transition directly into stable, well-paying jobs in the region’s key industries.

2. Strengthen Regional Postsecondary Pathways

Graduates who complete college within Northeast Minnesota are far more likely to work in the region.

Expanding access to — and awareness of — local postsecondary options can help more students see higher education as part of their home region’s opportunity structure.

Action steps:

- **Enhance collaboration between regional colleges and employers** to align degree programs with emerging labor market needs.
- **Expand transfer and articulation agreements** between two-year and four-year institutions within the region to improve mobility without relocation.
- **Create financial or housing incentives** for students who attend college locally, especially in high-demand career fields.

Policy impact: Regional colleges become both educational anchors and economic engines, helping the region retain its own talent.

3. Balance Academic Mobility with Regional Connection

High academic achievement and college readiness open doors — but they often lead graduates away from the region. Mobility itself is not a problem; losing connection is.

The region can celebrate success while building systems that make it easier for mobile graduates to return, contribute, or stay engaged.

Action steps:

- **Develop a “Return to Region” initiative** offering incentives for graduates to come back after college — such as student loan forgiveness, relocation stipends, or targeted job placement.
- **Leverage alumni networks and virtual engagement** to connect mobile graduates with local opportunities.
- **Expand hybrid and remote work options** with local employers to tap into talent living elsewhere.

Policy impact: These strategies help convert geographic mobility into long-term regional benefit.

4. Reconnect the Unanchored Pathway

The Unanchored Pathway represents one of the greatest untapped opportunities in the region’s labor market. Many of these young adults are willing and able to work but remain outside stable employment due to structural and socioeconomic barriers.

Action steps:

- **Expand re-engagement programs** that combine education, training, and wraparound support for non-college youth.
- **Use CTE and adult education as re-entry points** for young adults who left school or college without credentials.
- **Integrate workforce development with social services** to address childcare, transportation, and housing barriers that limit participation.

Policy impact: Reconnecting even a fraction of these individuals would not only improve lives but significantly strengthen the region’s labor supply.

5. Improve Data Coordination and Tracking

Understanding graduate outcomes across regions — especially for those working outside Minnesota — remains a persistent challenge.

Better data systems can help policymakers distinguish between disconnection and mobility, leading to more targeted and effective interventions.

Action steps:

- **Expand data-sharing agreements** between Minnesota and neighboring states to capture out-of-state employment outcomes.
- **Coordinate education, workforce, and economic development data systems** to track students from high school through long-term employment.
- **Use this information to inform program design and funding priorities** at both regional and state levels.

Policy impact: Clearer data ensures smarter investments — allowing Minnesota to measure not just how many graduates succeed, but *where and how* they do.

6. Embed Equity into Workforce Strategies

The data show that students from **low-income backgrounds** and those who **received special education services** face persistent barriers to postsecondary completion and meaningful work. Yet the same data also show that targeted supports — especially CTE participation — can close those gaps.

Action steps:

- **Integrate equity goals into CTE and workforce programs**, ensuring that low-income and special education students have equal access to high-quality training and support.
- **Provide targeted scholarships and stipends** for underrepresented students pursuing high-demand regional programs.
- **Train educators and employers** to recognize and address systemic barriers to participation.

Policy impact: Equity-focused investments strengthen both fairness and the regional workforce by expanding opportunity to those most likely to stay and contribute.

7. Foster a Shared Regional Narrative

Finally, the data reveal that perception matters. Many young people — especially high achievers — leave the region not because opportunities don't exist, but because they are unaware of them or assume better prospects lie elsewhere. Changing that narrative requires coordination among schools, colleges, employers, and community leaders.

Action steps:

- **Promote awareness of local career and education opportunities** through coordinated regional marketing campaigns.
- **Highlight success stories** of graduates who have built fulfilling careers locally.
- **Engage employers directly with high schools** to demonstrate career options early.

Policy impact: A stronger regional identity helps ensure that “success” includes staying, returning, or contributing to Northeast Minnesota’s economy.

Outlook

The story of Northeast Minnesota’s graduates is not one of loss — it’s one of potential.

The region already has the tools it needs to strengthen each pathway: strong CTE programs, resilient local colleges, committed educators, and communities eager to keep their youth close to home. What’s needed now is **alignment and investment** — bringing education, workforce, and economic systems together around a shared goal: ensuring that every graduate, regardless of their path, has the opportunity to build a meaningful life connected to this region.

When local learning leads to local opportunity, everyone benefits — individuals, employers, and the communities that sustain them.